Lunar localization: what limits the current matcher?

Development diagnostic | Thomas de Chillaz | 31 October 2026

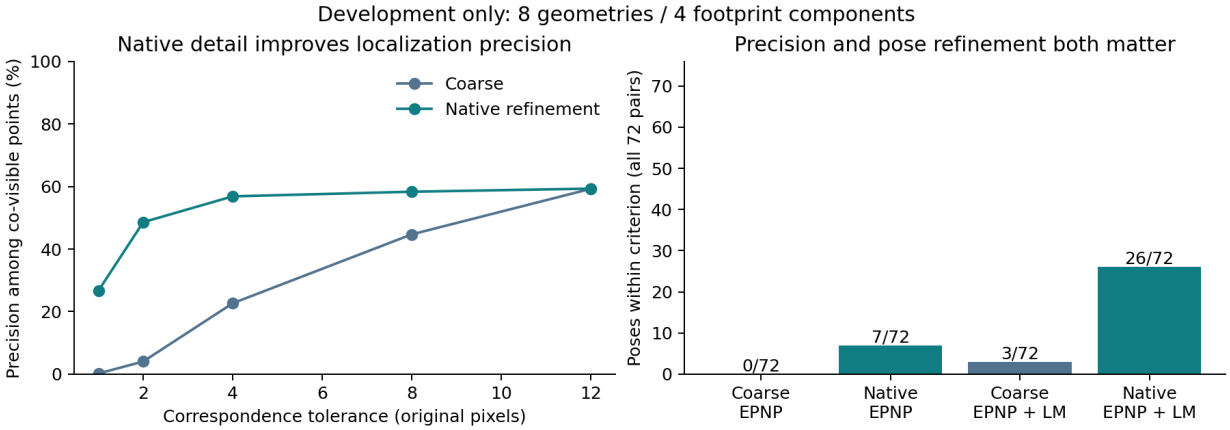
Result: recovering useful camera pose needs finer correspondences and a well-refined pose estimate. Standard engineering changes recover 26/72 development poses. A new learned compact-v3 model has not been trained.

Question and protocol

Eight development stereo geometries, four selected footprint components, three lighting observations per camera: 72 dependent stereo/light combinations. Exact ray-depth and calibrated cameras define the simulated reference. Query depth and true query pose enter scoring only. Pose success requires rotation error at most 2 degrees and translation error at most 1% of median reference range.

Matcher frontend	Original pose fit	With LM refit
Frozen coarse matcher	0/72	3/72
Native descriptor refinement	7/72	26/72

The original-estimator experiment was specified before inference. LM is a separately disclosed post-diagnostic development ablation: standard nonlinear optimization on the original RANSAC inliers, without changing the 3-pixel threshold. No old result or model was adjusted.

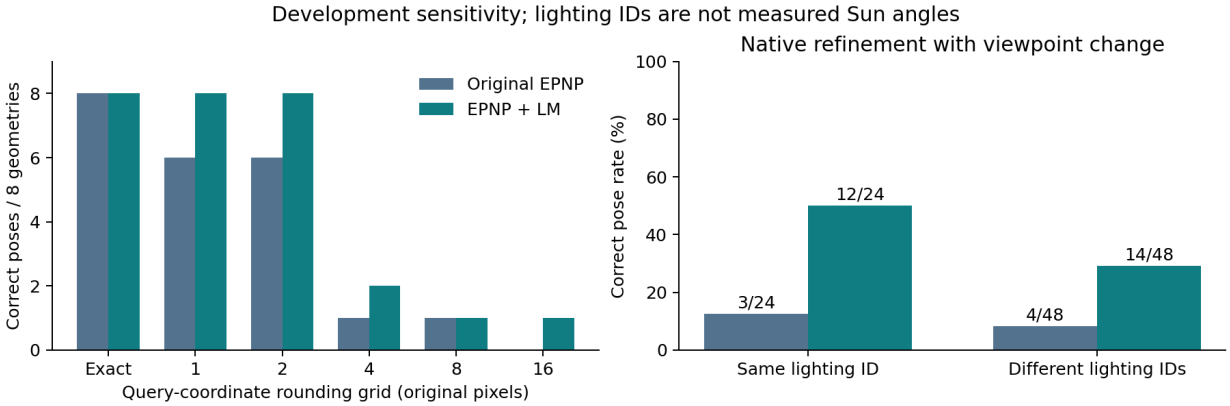


Limits: only 1/72 refined predictions meets the stricter 0.5-degree / 0.1%-range criterion. These are raw pose outcomes, without a calibrated rejection policy. They establish neither reliable navigation nor state-of-the-art superiority.

What the diagnosis establishes

Precision matters: exact correspondences recover 8/8 poses. Rounding correct query locations to a 16-original-pixel grid gives 0/8 with the original pose fit. Native descriptor refinement raises correspondence precision within 2 original pixels from 4.06% to 48.57% among co-visible points.

Both frontends return 22,697 matches; 14,938 reference points are co-visible. Native refinement plus LM succeeds on 12/24 same-light and 14/48 changing-light pairs. Lighting IDs are not measured Sun angles. LM is not universally better: it loses cases at some noise levels. Performance remains too weak for a navigation claim.



Fresh-data gate: failed before training

All 76,802 released SouthPole camera files were inspected: 12,974 stereo geometries, with only intrinsics and camera-to-world matrices. No trajectory or generation-batch IDs are released. The advertised Tycho directory is empty. Conservatively retaining entire mode/folders leaves zero fresh fallback groups; 12 are required. This does not mean all unused terrain is exhausted; the requested split cannot be certified from the available metadata.

Consequently, training, modern pose rankings, transfer evaluation and a superiority paper remain deferred. Verified batch/trajectory grouping plus untouched spatial data, or an additional suitable 3D relighting source, is needed before the learned-method phase.

Reproducibility and next comparison

Saved correspondences, native coordinates, protocols and the full camera inventory are checksummed. Independent replay checks 312 diagnostic rows and 15 tables, plus 312 refit poses and 15 tables. Repository checks: 137 tests passed, four optional tests skipped; fresh procedural smoke passed. Native feature-extraction costs are reported separately; no end-to-end speed claim is made.

Next: train pair-specific reliability and native refinement on true 3D correspondences; compare against deterministic refinement plus pose refitting, RoMa v2, LoMa-R, EfficientLoFTR and lunar-adapted MOONSt3R. Real orbital navigation accuracy remains unverified.

Sources: MoonAnything, doi:10.5281/zenodo.18415119; Rothenberger et al., doi:10.2514/6.2025-2073; RoMa v2, arXiv:2511.15706. Full methods, caveats and commands are in the repository diagnostic report.